ANALISIS PCA

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PENGERTIAN

Analisis PCA (*Principal Component Anaylsis*) adalah teknik statistika dan machine learning (unsupervised) untuk mengurangi dimensi data yang kompleks menjadi sedikit variabel baru/komponen utama yang lebih sederhana, namun tetap mempertahankan sebagian besar varians atau informasi dari data asli. Tujuannya adalah untuk mengurangi jumlah variabel tanpa kehilangan banyak informasi, memudahkan visualisasi data, dan mempercepat pemrosesan model.

DATASET YANG DIGUNAKAN

Dataset yang digunakan untuk analisis PCA adalah dataset "Student Performance Prediction" di Kaggle. Dataset ini berisi informasi mengenai prestasi akademik siswa dari dua sekolah menengah di Portugal (Sekolah Gabriel Pereira dan Mousinho da Silveira). Data ini menggabungkan nilai akademik, demografi siswa, dan kebiasaan sosial mereka. Dalam dataset ini, terdapat 649 siswa (baris) dan 34 variabel dengan 16 variabel tersebut adalah numerik.

ANALISIS PCA

1. Persiapkan data tersebut ke R

```{r}
# Persiapan Data library di R
library(readxl)
library(REdaS)
```

```{r}
# Load Dataset
student_performance <- read_excel("C:/Users/Benhard Leroy/Downloads/Tugas 2 Komputasi Statistika.xlsx",
sheet = "student_performance")
student_performance
str(student_performance)
```

age <dbl>	Medu <dbl>	Fedu <dbl>	traveltime <dbl>	studytime <dbl>	failures <dbl>	famrel <dbl>	freetime <dbl>	goout <dbl>	Dalc <dbl>
18	4	4	2	2	0	4	3	4	1
17	1	1	1	2	0	5	3	3	1
15	1	1	1	2	3	4	3	2	2
15	4	2	1	3	0	3	2	2	1
16	3	3	1	2	0	4	3	2	1
16	4	3	1	2	0	5	4	2	1
16	2	2	1	2	0	4	4	4	1
17	4	4	2	2	0	4	1	4	1
15	3	2	1	2	0	4	2	2	1
15	3	4	1	2	0	5	5	1	1

1-10 of 395 rows | 1-10 of 16 columns

Previous 1 2 3 4 5 6 ... 40 Next

ANALISIS PCA

```
tibble [395 × 16] (S3: tbl_df/tbl/data.frame)
 $ age       : num [1:395] 18 17 15 15 16 16 16 17 15 15 ...
 $ Medu      : num [1:395] 4 1 1 4 3 4 2 4 3 3 ...
 $ Fedu      : num [1:395] 4 1 1 2 3 3 2 4 2 4 ...
 $ traveltime: num [1:395] 2 1 1 1 1 1 1 2 1 1 ...
 $ studytime : num [1:395] 2 2 2 3 2 2 2 2 2 2 ...
 $ failures  : num [1:395] 0 0 3 0 0 0 0 0 0 0 ...
 $ famrel    : num [1:395] 4 5 4 3 4 5 4 4 4 5 ...
 $ freetime  : num [1:395] 3 3 3 2 3 4 4 1 2 5 ...
 $ goout     : num [1:395] 4 3 2 2 2 2 4 4 2 1 ...
 $ Dalc      : num [1:395] 1 1 2 1 1 1 1 1 1 1 ...
 $ Walc      : num [1:395] 1 1 3 1 2 2 1 1 1 1 ...
 $ health    : num [1:395] 3 3 3 5 5 5 3 1 1 5 ...
 $ absences  : num [1:395] 6 4 10 2 4 10 0 6 0 0 ...
 $ G1        : num [1:395] 5 5 7 15 6 15 12 6 16 14 ...
 $ G2        : num [1:395] 6 5 8 14 10 15 12 5 18 15 ...
 $ G3        : num [1:395] 6 6 10 15 10 15 11 6 19 15 ...
```

ANALISIS PCA

2. Hitung Statistika Deskriptif

```{r}
# Statistik Deskriptif
summary(student_performance)
```

```
      age            Medu            Fedu          traveltime       studytime        failures      
 Min.   :15.0   Min.   :0.000   Min.   :0.000   Min.   :1.000   Min.   :1.000   Min.   :0.0000  
 1st Qu.:16.0   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:0.0000  
 Median :17.0   Median :3.000   Median :2.000   Median :1.000   Median :2.000   Median :0.0000  
 Mean   :16.7   Mean   :2.749   Mean   :2.522   Mean   :1.448   Mean   :2.035   Mean   :0.3342  
 3rd Qu.:18.0   3rd Qu.:4.000   3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:0.0000  
 Max.   :22.0   Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :3.0000  
     famrel         freetime         goout           Dalc            Walc           health      
 Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000  
 1st Qu.:4.000   1st Qu.:3.000   1st Qu.:2.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:3.000  
 Median :4.000   Median :3.000   Median :3.000   Median :1.000   Median :2.000   Median :4.000  
 Mean   :3.944   Mean   :3.235   Mean   :3.109   Mean   :1.481   Mean   :2.291   Mean   :3.554  
 3rd Qu.:5.000   3rd Qu.:4.000   3rd Qu.:4.000   3rd Qu.:2.000   3rd Qu.:3.000   3rd Qu.:5.000  
 Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000   Max.   :5.000  
    absences           G1              G2              G3      
 Min.   : 0.000   Min.   : 3.00   Min.   : 0.00   Min.   : 0.00  
 1st Qu.: 0.000   1st Qu.: 8.00   1st Qu.: 9.00   1st Qu.: 8.00  
 Median : 4.000   Median :11.00   Median :11.00   Median :11.00  
 Mean   : 5.709   Mean   :10.91   Mean   :10.71   Mean   :10.42  
 3rd Qu.: 8.000   3rd Qu.:13.00   3rd Qu.:13.00   3rd Qu.:14.00  
 Max.   :75.000   Max.   :19.00   Max.   :19.00   Max.   :20.00  
```

```{r}
# Cek Missing Value
student_performance_misval <- na.omit(student_performance)
student_performance_misval
```

A tibble: 395 x 16

age <dbl>	Medu <dbl>	Fedu <dbl>	traveltime <dbl>	studytime <dbl>	failures <dbl>	famrel <dbl>	freetime <dbl>	goout <dbl>	Dalc <dbl>
18	4	4	2	2	0	4	3	4	1
17	1	1	1	2	0	5	3	3	1
15	1	1	1	2	3	4	3	2	2
15	4	2	1	3	0	3	2	2	1
16	3	3	1	2	0	4	3	2	1
16	4	3	1	2	0	5	4	2	1
16	2	2	1	2	0	4	4	4	1
17	4	4	2	2	0	4	1	4	1
15	3	2	1	2	0	4	2	2	1
15	3	4	1	2	0	5	5	1	1

1-10 of 395 rows | 1-10 of 16 columns

Previous 1 2 3 4 5 6 ... 40 Next

ANALISIS PCA

3. Hasil uji Bartlett dan KMO

```{r}
# Uji Bartlett
uji_bartlett <- bartlett.test(student_performance)
uji_bartlett
```

```
        Bartlett test of homogeneity of variances

data:  student_performance
Bartlett's K-squared = 8002.4, df = 15, p-value < 2.2e-16
```

```{r}
# Uji KMO
uji_kmo <- KMOS(student_performance)
uji_kmo
```

```
Kaiser-Meyer-Olkin Statistics

Call: KMOS(x = student_performance)

Measures of Sampling Adequacy (MSA):
       age       Medu       Fedu traveltime  studytime   failures     famrel   freetime      goout
 0.6341464  0.6192996  0.5980620  0.7292107  0.7165847  0.8853567  0.3741483  0.5673121  0.6523210
      Dalc       Walc     health   absences         G1         G2         G3
 0.6313317  0.5968986  0.5463773  0.4370603  0.8347685  0.6985535  0.7391452

KMO-Criterion: 0.6913068
```

ANALISIS PCA

4. Analisis PCA (a. Simpan output dan tampilkan komponen output)

```{r}
# Analisis PCA

# a. Simpan Output dan tampilkan komponen output
student_performance_pca <- prcomp(student_performance, center = TRUE, scale = TRUE, retx = TRUE)
names(student_performance_pca)
```

```
[1] "sdev"     "rotation" "center"   "scale"    "x"
```

ANALISIS PCA

4. Analisis PCA (b. Tampilkan output PCA)

```{r}
# b. Tampilkan output PCA
student_performance_pca
```

```
Standard deviations (1, .., p=16):
 [1] 1.8427168 1.4464904 1.2480804 1.1312336 1.0814941 1.0063491 0.9646386 0.9478895 0.8774575
[10] 0.8467817 0.7962918 0.7869959 0.6033879 0.5431673 0.4285826 0.2829687

Rotation (n x k) = (16 x 16):
                   PC1          PC2          PC3          PC4          PC5         PC6
age        -0.16559335 -0.067118310 -0.251968900 -0.176948586  0.559304880  0.13190885
Medu        0.23005821 -0.212822178  0.536135590 -0.112648183  0.100740462  0.01861273
Fedu        0.20760118 -0.197155061  0.564329289 -0.082110225 -0.007483045  0.05763956
traveltime -0.14548670 -0.036313008 -0.242593268 -0.062458145 -0.265704660 -0.11389394
studytime   0.15587638  0.216796572  0.006639562 -0.015736764  0.334702356 -0.31868916
failures   -0.31776551 -0.003337548 -0.129664826 -0.021856702  0.117378020  0.16124295
famrel      0.01946917  0.025529960  0.013926599  0.553305720  0.384957096  0.23065662
freetime   -0.06417520 -0.290235950 -0.008714771  0.515115969  0.136375975 -0.10965828
goout      -0.16588791 -0.374682148  0.100793686  0.163130484  0.267376820 -0.35034435
Dalc       -0.16719845 -0.500409666 -0.090725308 -0.089018215 -0.134066648 -0.04464123
Walc       -0.19140012 -0.518703688 -0.099680300 -0.110959342 -0.146549746 -0.07334940
health     -0.07038987 -0.058477770  0.061954353  0.347724474 -0.293080808  0.62333921
absences   -0.03737120 -0.159012099  0.007890225 -0.451767422  0.338660952  0.49617661
G1          0.45106642 -0.150333967 -0.267697024  0.024033233  0.041781047  0.03183674
G2          0.46776230 -0.170127627 -0.276543717 -0.003956968 -0.019494984  0.01970855
G3          0.45265396 -0.194972436 -0.272505011  0.026311469 -0.011244584  0.08313152
```

```
                   PC7          PC8         PC9         PC10          PC11         PC12
age        0.091587116  0.001940761  0.49137307 -0.275548423  0.2052199418  0.39111878
Medu       0.094458638 -0.104135951  0.20885378  0.110669762  0.0363269985 -0.15029096
Fedu       0.125174560 -0.108485356  0.29046113 -0.105290024 -0.0159233486  0.01045134
traveltime 0.732726725 -0.355822819  0.19638619  0.346349676 -0.0707096521  0.01545633
studytime  0.271407600  0.644470839  0.05670219  0.327650348  0.1445160539 -0.25773672
failures  -0.312944832 -0.086759766  0.44803305  0.189453530 -0.3546354355 -0.60631431
famrel     0.384193465 -0.174573060 -0.25545981 -0.353335756 -0.0507674443 -0.33820289
freetime  -0.273430345 -0.231574898  0.02975156  0.491682530  0.4424005750  0.12375283
goout      0.020170143  0.121784267 -0.10524839  0.096286837 -0.6221096261  0.29141211
Dalc       0.085596035  0.183703588 -0.01200592 -0.174509879  0.4170626742 -0.32825025
Walc       0.052186958  0.216082062 -0.13313008 -0.208721045 -0.0569270406 -0.07140381
health     0.120314421  0.492107215  0.20184678  0.155986809 -0.0959067832  0.22992254
absences   0.050142289 -0.071314511 -0.46813328  0.402347979  0.0004487468  0.00836378
G1         0.002505119 -0.019712727  0.16588632  0.028180740 -0.0653550919  0.03709191
G2        -0.087204246  0.004320121  0.06860251 -0.004974327 -0.1011090597 -0.05973733
G3        -0.033305506 -0.029461507 -0.02904018  0.024737002 -0.1359041618 -0.06857809
```

ANALISIS PCA

4. Analisis PCA (b. Tampilkan output PCA)

	PC13	PC14	PC15	PC16
age	-0.066410861	0.068443308	-0.114321628	-0.005753982
Medu	-0.655666771	0.236235456	0.076799393	0.002859790
Fedu	0.659600336	-0.126624682	-0.122790824	-0.002345393
traveltime	0.005916867	0.018029969	-0.046259366	0.037801281
studytime	0.137431039	0.084391841	-0.042282412	-0.011696359
failures	0.077247095	0.009745347	0.009969174	-0.017708600
famrel	0.016814242	0.040068971	0.041164343	0.050060426
freetime	0.123455111	0.130748657	-0.030957183	0.015658074
goout	-0.092767695	-0.290756397	-0.017255176	0.011006820
Dalc	-0.133279998	-0.559608943	-0.023633202	-0.025836529
Walc	0.185270226	0.692831260	0.107990657	0.012765625
health	-0.083257509	-0.031550692	-0.013984359	0.022857164
absences	0.107063014	-0.062094056	0.043383207	0.048753713
G1	0.091741858	-0.107655965	0.777516051	-0.200120671
G2	-0.019649185	-0.013069494	-0.218884956	0.774842597
G3	-0.048097421	0.053883219	-0.540589601	-0.592488236

ANALISIS PCA

4. Analisis PCA (c. Tampilkan summary dari output PCA)

```{r}
# c. Tampilkan summary dari output PCA
summary(student_performance_pca)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4    PC5    PC6     PC7     PC8     PC9    PC10
Standard deviation     1.8427 1.4465 1.24808 1.13123 1.0815 1.0063 0.96464 0.94789 0.87746 0.84678
Proportion of Variance 0.2122 0.1308 0.09736 0.07998 0.0731 0.0633 0.05816 0.05616 0.04812 0.04481
Cumulative Proportion  0.2122 0.3430 0.44035 0.52033 0.5934 0.6567 0.71489 0.77105 0.81917 0.86398
                          PC11    PC12    PC13    PC14    PC15  PC16
Standard deviation     0.79629 0.78700 0.60339 0.54317 0.42858 0.283
Proportion of Variance 0.03963 0.03871 0.02275 0.01844 0.01148 0.005
Cumulative Proportion  0.90361 0.94232 0.96508 0.98352 0.99500 1.000
```

ANALISIS PCA

3. Melakukan Pembersihan Data

```
   student_id  age gender academic_level  study_hours  self_study_hours  \
0           1   18  Other    High School         7.64              1.56
1           2   18  Other    High School         2.21              2.22
2           3   22   Male    High School         3.45              0.00
3           4   17  Other    High School         5.75              2.08
4           5   19  Other    High School         6.83              1.72

   online_classes_hours  social_media_hours  gaming_hours  sleep_hours  ...  \
0                  2.20                3.05          2.19         6.52  ...
1                  2.10                1.65          2.55         5.97  ...
2                  0.29                1.34          2.08         8.39  ...
3                  3.01                2.27          2.20         6.31  ...
4                  3.33                2.65          0.70         8.01  ...

   exercise_minutes  caffeine_intake_mg  part_time_job  upcoming_deadline  \
0                81                  38              1                  0
1               111                 339              0                  0
2                68                 266              0                  1
3               113                 480              1                  1
4               121                  24              1                  0

   internet_quality mental_health_score  focus_index  burnout_level  \
0              Good                  10        43.05          31.77
1              Good                   3        15.92          37.00
2              Good                   8        27.39          34.37
3              Poor                   3        22.31          77.31
4              Good                   8        38.11          39.53
```

```
   productivity_score  exam_score
0               73.65       50.16
1               13.70        1.00
2               45.15       18.30
3               20.92        9.37
4               59.23       27.81

[5 rows x 21 columns]
```

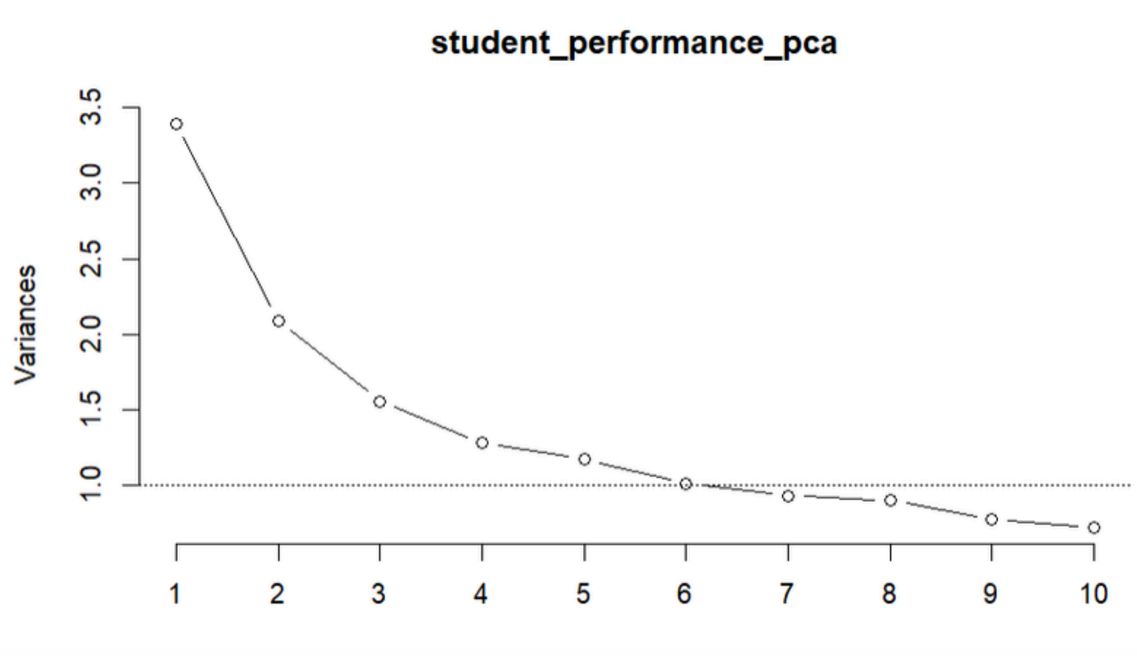
ANALISIS PCA

5. Melakukan Visualisasi Data dari output PCA

```{r}
# Melakukan Visualisasi Data dari output PCA
screeplot(student_performance_pca,type = "line")
abline(h=1, col = "black", lty = 3)

biplot(student_performance_pca, scale = 0)
```

Output Screeplot output PCA



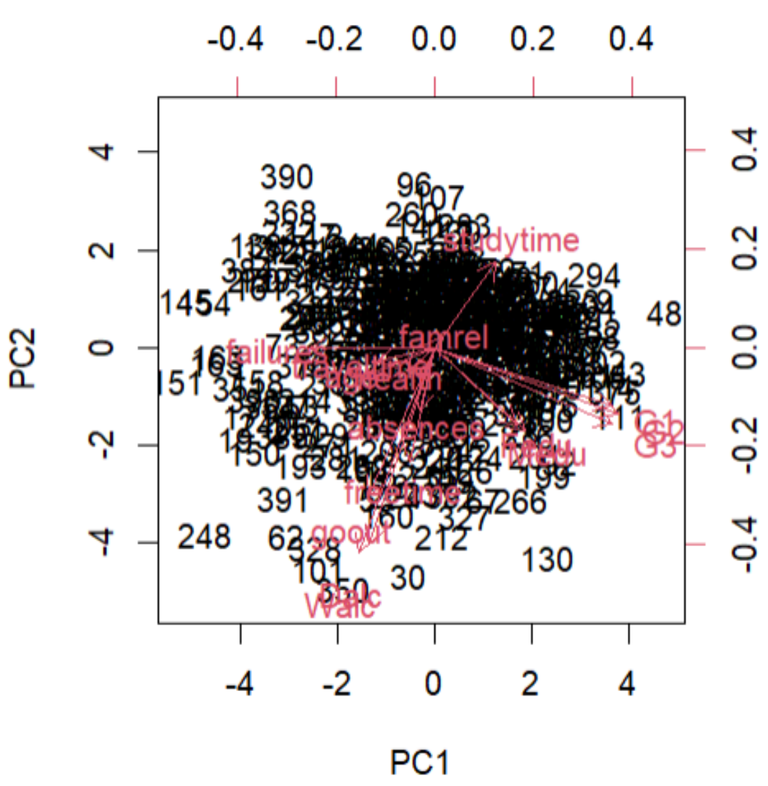
ANALISIS PCA

5. Melakukan Visualisasi Data dari output PCA

```{r}
# Melakukan Visualisasi Data dari output PCA
screeplot(student_performance_pca,type = "line")
abline(h=1, col = "black", lty = 3)

biplot(student_performance_pca, scale = 0)
```

Output biplot output PCA



KESIMPULAN

Dari hasil output dari Student Performance Dataset dari Kaggle, dapat dilihat proporsi proporsi variansi yang dijelaskan oleh masing-masing komponen utama:

1. Komponen 1 (PC1) menjelaskan proporsi variansi terbesar.
2. Komponen 2 (PC2) menjelaskan proporsi variansi terbesar kedua, dan seterusnya.

Dalam screeplot tersebut, kita menunjukkan adanya penurunan eigenvalue dari masing-masing komponen semakin kecil variansnya. Selanjutna dari biplot, kita menunjukkan hubungan antara variabel asli dengan komponen utama.

Dari biplot dapat dilihat:
a. Variabel-variabel yang berdekatan menunjukkan korelasi positif.
b. Variabel yang berlawanan arah menunjukkan korelasi negatif.
c. Panjang vektor menunjukkan seberapa besar kontribusi variabel terhadap komponen utama.

Secara keseluruhan, dilihat bahwa analisis PCA berhasil mereduksi dimensi data dan menunjukkan bahwa sebagian besar informasi dalam data dapat dijelaskan oleh dua atau tiga komponen utama pertama, dimana hasil tersebut itu sangat berguna untuk analisis lanjutan.

SUMBER

- Rpubs principal component analysis

TERIMA KASIH